

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical and Automation Engineering) SEMESTER - 3 Winter 2025 (Regular)

Course :Bachelor of Technology (Mechanical and Automation Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AF10THOEM05A - INTRODUCTION TO AUTOMATION

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12**
- 1 Automobile assembly lines are an example of:
A. Job shop production B. Batch production
C. Mass production D. Continuous production
 - 2 The highest level in automation architecture is:
A. Enterprise level B. Supervisory level
C. Control level D. Field level
 - 3 Sensors belong to which level of the automation pyramid?
A. Level 3 B. Level 2
C. Level 1 D. Level 0
 - 4 Actuators are used to:
A. Sense parameters B. Process data
C. Perform physical action D. Store information
 - 5 Pneumatic systems mainly use:
A. Oil B. Compressed air
C. Water D. Gasoline
 - 6 A direction control valve is used to:
A. Store energy B. Convert energy
C. Control pressure D. Control the direction of fluid flow
 - 7 Feedforward control acts:
A. After error occurs B. Before disturbance affects output
C. Without sensors D. Only in manual systems
 - 8 Which device provides interaction between humans and machines?
A. PLC B. Sensor
C. HMI D. Actuator
 - 9 PLC stands for:
A. Programmable Logic Controller B. Process Logic Computer
C. Programmable Line Control D. Process Level Controller

- 10 SCADA stands for:
 A. System Control and Data Acquisition
 B. Supervisory Control and Data Acquisition
 C. Smart Control and Data Automation
 D. Software Control and Data Analysis
- 11 Which learning type is related to machine learning?
 A. Supervised learning
 B. Unsupervised learning
 C. Reinforcement learning
 D. All of the above
- 12 Machine vision is a technology that uses
 A. Valve
 B. oil
 C. Camera
 D. water
- Q2. Solve the following.
- A) Differentiate between Industrial Automation and Industrial Information Technology 6
- B) Explain with neat sketch, types of automation systems. 6
- Q3. Solve the following.
- A) Write note on Industrial Sensors and Instrument Systems. 6
- B) List the five levels of the industrial Automation Pyramid and briefly describe the primary function of each level. 6
- Q4. Solve Any Two of the following.
- A) Differentiate between hydraulics and pneumatics. 6
- B) Explain with neat sketch working of Supervisory Control system 6
- C) Explain the fundamental difference between a Continuous Control system and a Sequence/ Logic Control system 6
- Q5. Solve Any Two of the following.
- A) What are the important characteristics of a Process 6
- B) Discuss any one process modelling example. 6
- C) Differentiate between Feedback and feed forward control system 6
- Q6. Solve Any Two of the following.
- A) Explain the main components of a PLC with block diagram and their functions. 6
- B) Explain supervised, unsupervised, and reinforcement learning. 6
- C) Define machine vision. Explain its working with a block diagram. 6

***** End *****